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A lightweight and cross-scale attention network for geological hazard detection in rescue robotics

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Accurate and efficient identification of hazard targets is crucial for enhancing the operational autonomy of rescue robots and ensuring human safety during geological hazard rescue operations. However, existing detection models struggle to simultaneously meet the requirements of high accuracy and lightweight design when addressing the challenges of extreme scale heterogeneity in geological hazard targets, weak discriminative features, and complex background interference. This limits their deployment on resource-constrained edge devices such as rescue robots. To address this issue, this paper proposes a lightweight Depthwise Separable Selective Kernel Attention (DSSKA) module, and based on it, constructs the DSSKA-YOLOv8n detection model. By employing multi-scale depthwise separable convolutions and an adaptive weight fusion mechanism, the proposed module significantly reduces the number of parameters while enhancing the perception and discrimination capabilities for multi-scale targets with weakly discriminative features. Experimental results on a self-constructed geological hazard target detection dataset demonstrate that compared to the baseline YOLOv8n model, our model achieves a 3.1% improvement in precision and a 2.9% increase in mAP50-95. Furthermore, compared to a counterpart model with the standard SKA (Selective Kernel Attention) module directly embedded, our model reduces the number of parameters by 61.2%, effectively achieving a synergistic optimization of accuracy and lightweight design. This study provides a promising lightweight algorithmic solution for the visual perception systems of rescue robots, laying the groundwork for subsequent deployment optimization on robot platforms. It holds positive implications for advancing the intelligent development of geological hazard emergency response.

Keywords YOLOv8n, DSSKA, Lightweight design, Geological hazard detection, Rescue robots

Geological hazards pose significant threats to both human safety and ecological stability globally, severely constraining regional sustainable development and posing an ongoing risk to human life and property[1]. Driven by the dual factors of global climate change and intensified engineering activities, the frequency and destructive intensity of primary geological hazards such as landslides and rockfalls are continuously increasing. Concurrently, secondary hazards including fallen trees and road collapses are becoming increasingly prominent[2]. Traditional methods of manual surveying and remote sensing interpretation are constrained by low efficiency, high rates of missed detections, and environmental limitations, making them inadequate for meeting the urgent demand for high-efficiency monitoring in post-hazard rescue operations. Consequently, as the core sensing platform in field hazard scenarios, rescue robots must possess the capability for accurate identification of multi-type and multi-scale geological hazard targets. This capability has become a critical technological prerequisite for reducing rescue risks and ensuring personnel safety[3].

With the rapid development of deep learning technologies, target detection algorithms, particularly the YOLO series, have emerged as mainstream solutions for the onboard perception of rescue robots due to their effective balance between efficiency and accuracy[4]. To address the specific challenges of geological hazard identification, researchers have conducted various improvements based on existing algorithms, primarily focusing on enhancing accuracy in complex scenarios, model lightweighting, multi-source data fusion, and innovative framework design. For instance, to improving detection accuracy, Wang et al.[5] proposed the YOLOv8-DBW model, which enhanced landslide detection capability by incorporating DySample upsampling, the BiFormer attention mechanism, and the WIoU (Weighted Intersection over Union) loss function. Li et al.[6]

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focused on the characteristics of old landslides in the Sichuan-Tibet region, integrating the WIoU loss with the CBAM (Convolutional Block Attention Module) attention mechanism, resulting in a 10.9% improvement in the detection accuracy of the YOLOv8-CW model. Pan et al.[7] proposed the Faster-Swin model, which improved landslide detection accuracy on low-quality remote sensing datasets by integrating a windowed multi-scale attention mechanism. In the realm of lightweighting and efficiency optimization, Du et al.[8] proposed the LBE-YOLO model, which significantly reduced computational load while improving accuracy through GhostConv and GhostC2f. modules. Lu et al.[9] constructed MS2LandsNet, achieving efficient detection via MFF (Multi-scale Feature Fusion) and a MSCA (Multi-Scale Channel Attention) mechanism. Regarding the fusion of multi-source data and innovative methodological frameworks, Mao et al.[10] combined InSAR technology with an improved YOLOv8, achieving an mAP50 of 93.4% in identifying potential landslides on the Tibetan Plateau. Yang et al.[11] integrated UAV LiDAR (Unmanned Aerial Vehicle Light Detection and Ranging), SBAS-InSAR (Small Baseline Subset Interferometric Synthetic Aperture Radar), and optical remote sensing technologies to achieve comprehensive identification of cascading hazards such as landslides, collapses, and debris flows in high-altitude valley regions. Furthermore, some studies have explored combinations with traditional algorithms or change detection. For example, He et al.[12] developed a model combining CNN (Convolutional Neural Network) with the Harris Hawks optimization algorithm and the Apriori algorithm, while Shi et al.[13] constructed a CDCNN (Change Detection Convolutional Neural Network) model by integrating deep CNN with change detection, both of which improved the accuracy and efficiency in large-scale landslide identification.

Despite their commendable performance in general tasks, existing methods often fall short in geological hazard identification, which is inherently a problem of specific target detection under complex scene constraints. Existing methods have yet to collaboratively address its three core challenges. Specifically: First, in tackling the extreme heterogeneity of target scales, existing methods often rely on fixed multi-scale architectures, lacking the ability to dynamically adjust the receptive field based on the actual size of targets. This results in insufficient adaptability to targets with extreme scales[5,7]. Second, regarding the challenge of weak feature discriminability caused by similarities between targets and backgrounds or partial occlusion, while attention mechanisms have been employed to enhance features, there is a lack of targeted enhancement specifically for the weakly discriminative features characteristic of geological hazards. Consequently, models remain susceptible to interference in complex scenes[6,11]. Third, when confronting the contradiction between harsh field environments and the stringent resource constraints of edge platforms, existing solutions often present a dilemma: lightweight models frequently sacrifice robustness, whereas high-precision schemes, due to their computational complexity and reliance on multi-source data, struggle to meet the stringent deployment requirements for low latency and high efficiency at the edge[10,11]. Therefore, a detection model suitable for rescue robots must be capable of simultaneously achieving multi-scale perception, targeted enhancement of weak features, and truly efficient lightweight design.

Moreover, even advanced attention mechanisms designed to enhance model perception, such as the SKA (Selective Kernel Attention), face challenges due to their inherent structural design in terms of computational overhead and parameter size. This makes them difficult to deploy directly on rescue robot platforms with stringent efficiency requirements. To address this, this paper proposes a lightweight detection model for geological hazard rescue robots—DSSKA-YOLOv8n. The core contribution of this work is a novel lightweight attention solution, which is articulated in the following three aspects:

(1) We propose the Depthwise Separable Selective Kernel Attention (DSSKA) module. It replaces standard convolutions with depthwise separable convolutions to construct multi-scale branches. This design significantly reduces the module's complexity and parameter count while largely preserving its multi-scale feature extraction capability. Furthermore, by introducing an adaptive channel-weight fusion mechanism, the module dynamically enhances the response to critical features of geological hazards.
(2) We construct an efficiently integrated DSSKA-YOLOv8n detection framework. The DSSKA module is embedded into key nodes of the YOLOv8n backbone network in a plug-and-play manner. This design achieves global attention guidance from low-level to high-level features without disrupting the original feature pyramid structure, thereby significantly improving the model's discriminative power for multi-scale, ambiguous hazard targets and its overall robustness.
(3) We conduct systematic validation and in-depth analysis using a real-world geological hazard dataset. Experimental results demonstrate that the proposed model significantly outperforms the baseline model in terms of detection accuracy, while also exhibiting advantages in model size and computational complexity compared to other mainstream attention-enhanced models. The experiments further verify its balanced capability between accuracy and computational efficiency. This work provides a laboratory-validated and feasible algorithmic prototype for efficient visual perception in rescue robots, establishing a core foundation for subsequent deployment and optimization on real robotic platforms.

Results

Model training

The training process and convergence trends are shown in Fig. 1. The training losses, including the bounding box regression loss (box_loss), classification loss (cls_loss), and distribution focal loss (dfl_loss), all decreased rapidly and stabilized after approximately 300 epochs. This indicates effective model learning without overfitting, validating the stability afforded by the DSSKA module. Concurrently, the performance metrics on the validation set—including Precision, Recall, mAP50, and mAP50-95—continuously rose and then converged. The steady improvement in mAP50-95 directly reflects the enhanced comprehensive detection capability of the model for multi-scale geological hazard targets.

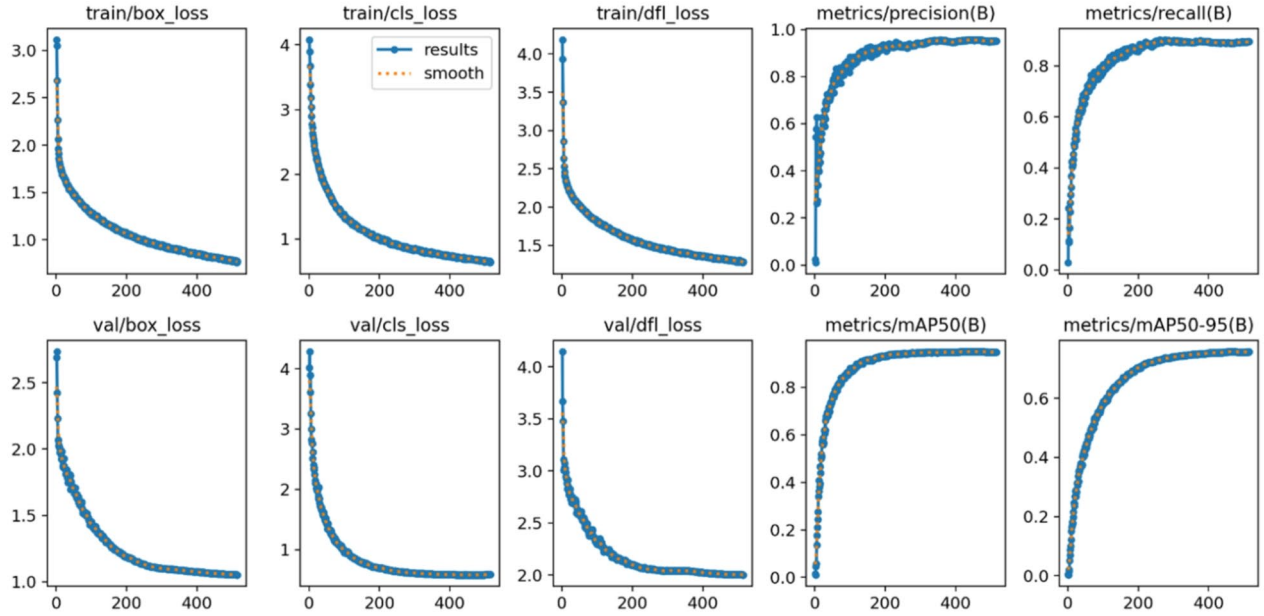


Fig. 1. Loss and metric curves of DSSKA-YOLOv8n.

Category	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)
ALL	95.5	88.9	84.9	75.8
Rockfall	94.3	84.3	92.5	72.1
Fallen_tree	97.1	84.2	92.2	70.8
Landslides	96.2	91.5	96.8	77.9
Road_collapse	94.5	95.6	98.2	82.3

Table 1. Detection performance of DSSKA-YOLOv8n.

The detailed performance of the model on the validation set is summarized in Table 1, where it achieves an overall mAP50-95 of 75.8%. Among the four hazard categories, "road_collapse" and "landslides"—characterized by more distinct structural features—attained the highest detection scores. In contrast, "rockfall" and "fallen_tree" yielded relatively lower performance. To investigate this discrepancy, we first examined the data distribution presented in Table 5. The results show that performance does not exhibit a simple positive correlation with sample size. For example, "rockfall" which possesses the largest number of samples, achieved the lowest mAP50-95, whereas "road_collapse" with the second-largest sample count, obtained the highest mAP50-95. Moreover, "fallen_tree" and "landslides" despite having comparable sample sizes, displayed markedly different detection outcomes. Consequently, class imbalance can be ruled out as the primary factor. We attribute the performance variation mainly to the intrinsic visual complexity of the targets. "road_collapse" typically presents regular shapes and stable textural patterns, which are easier for the model to learn and generalize. On the other hand, "rockfall" exhibits high heterogeneity in size, shape, surface texture, and background context; this substantial feature-level complexity increases learning difficulty and leads to suboptimal detection. The lower performance for "fallen_tree" may arise from its high visual similarity to natural backgrounds and limited diversity in the original sample features. These findings not only confirm the effectiveness of the DSSKA module in enhancing multi-scale features, but also highlight the inherent challenges of relying solely on visual perception for highly complex geological targets.

To determine the optimal decision threshold, Fig. 2 presents the curve of the F1-score variation with the confidence threshold. The model achieves the optimal comprehensive F1-score of 0.92 when the confidence threshold is approximately 0.537, balancing precision and recall. This curve provides a direct basis for selecting the confidence threshold in practical applications, further validating the model's usability on real-world scene data.

Ablation experiments for each part

To systematically validate the effectiveness of the proposed Depthwise Separable Selective Kernel Attention (DSSKA) module and determine its optimal deployment strategy, we designed a two-stage ablation study. All experiments were conducted under the same dataset and training configurations to ensure comparability of the results.

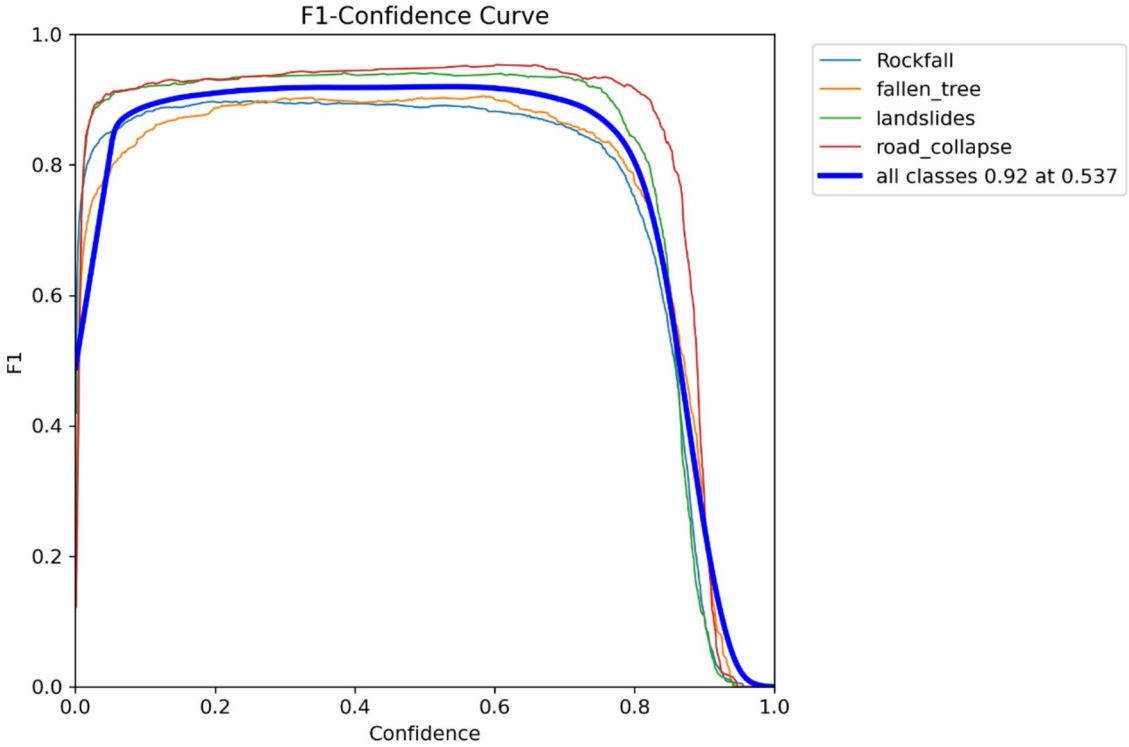


Fig. 2. F1-score vs. confidence threshold for each category.

Model	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)	Params /M
YOLOv8n	92.4	88.9	94.1	72.9	3.0
YOLOv8n-SKA	94.3	89.5	94.2	75.1	8.5
DSSKA-YOLOv8n	95.5	88.9	94.9	75.8	3.3

Table 2. Ablation study on the DSSKA module.

The first stage validated the core value of the DSSKA module. As shown in Table 2, introducing the standard Selective Kernel Attention (SKA) module to the baseline YOLOv8n model improved its overall detection accuracy (mAP50-95) by 2.2%, but at the cost of a substantial 183% increase in the model's parameter count. In contrast, incorporating the proposed lightweight DSSKA module achieved an even greater improvement in accuracy while increasing the parameter count by only 10%. This demonstrates that the DSSKA module can significantly enhance the model's feature perception capability at a minimal computational cost.

The second stage further investigated the impact of the module's placement when extending its integration to the feature fusion network (neck network). We designed four distinct architectural configurations (Model1-Model4), corresponding to the insertion of the module at key nodes before, during, and after deep feature fusion, as illustrated in Fig. 3. The corresponding quantitative evaluation results are listed in Table 3. The experimental results indicate that even the optimal deployment scheme within the neck network yields only marginal performance improvements while introducing additional computational overhead and latency. Based on these results from both stages, the strategy we ultimately adopted is to embed the DSSKA module solely into key stages of the backbone network. This scheme achieves the optimal balance between model complexity and detection accuracy, constituting the final and efficient architecture of the DSSKA-YOLOv8n model.

Comparative experiments for each model

To comprehensively evaluate DSSKA-YOLOv8n for geological hazard detection, we compared it with several representative models under identical dataset and training settings. The evaluated models included lightweight YOLO variants (v8n, v10n, v11n, v12n), the classic two-stage Faster R-CNN, the Transformer-based DETR, and the geohazard-specific YOLOv8n-CW. Detailed comparative results are summarized in Table 4.

As shown in Table 4, the proposed DSSKA-YOLOv8n achieves the best results across all core detection metrics, attaining a precision of 95.5% and an overall mAP50-95 of 75.8%. Compared to other lightweight models in the same series, our method achieves a significant improvement in accuracy with only a modest increase in parameter count. Specifically, its mAP50-95 is 2.4% higher than that of the second-best model, YOLOv11n,

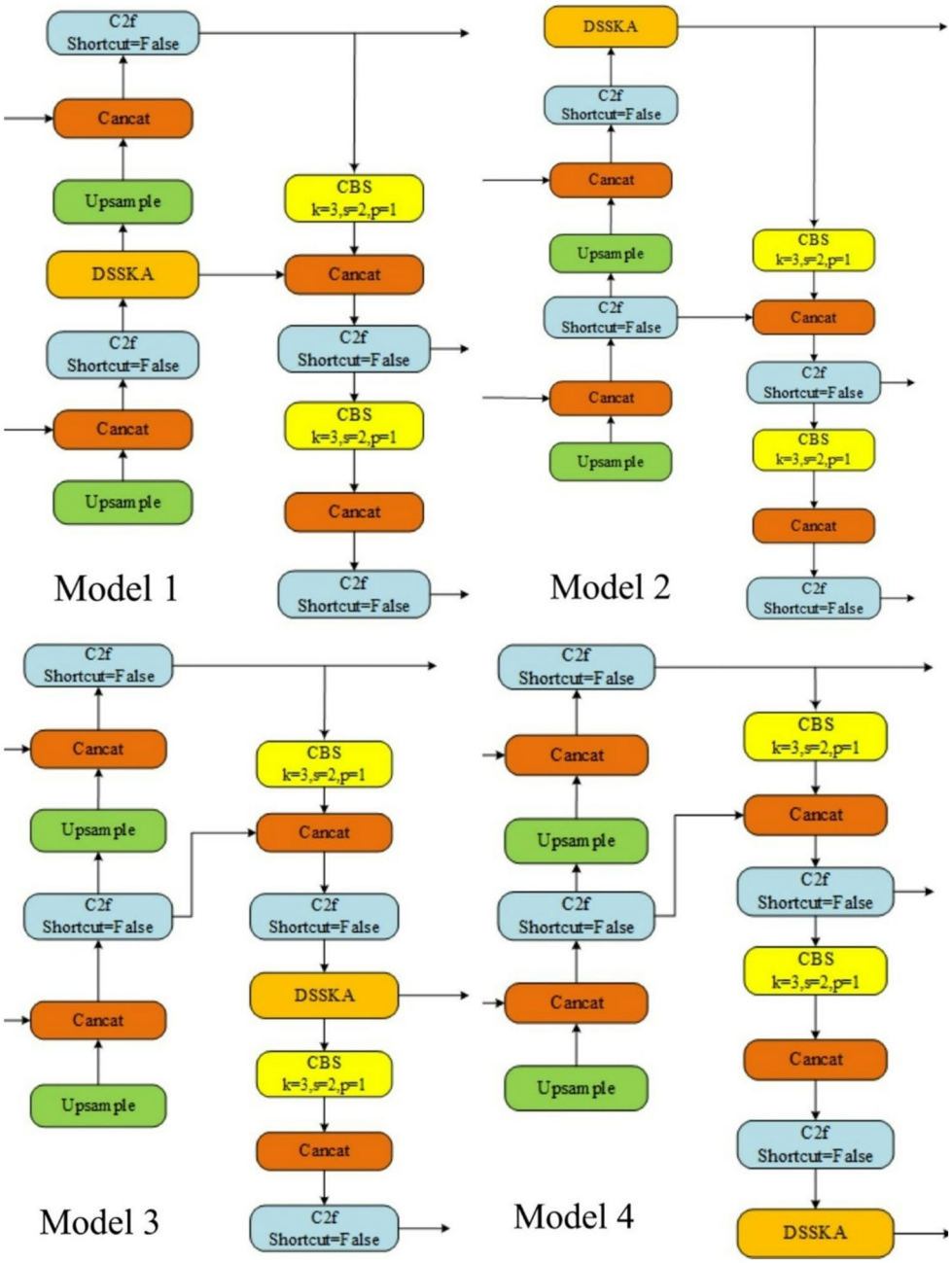


Fig. 3. Topological structure diagram of different DSSKA deployment locations.

Model	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)	Params /M
Model1	94.5	89.4	94.4	75.1	3.4
Model2	94.5	87.7	93.5	74.8	3.4
Model3	92.8	87.8	92.9	73.6	3.5
Model4	92.2	87.0	93.2	74.3	5.5

Table 3. Performance comparison of the ablation study on DSSKA with different deployment locations.

and it also significantly outperforms the YOLOv8n-CW model, which was specifically modified for geological hazard scenarios. Compared to heavyweight models like DETR and Faster R-CNN, our method achieves higher accuracy with only about one-tenth of their parameters, demonstrating its superior suitability for resource-constrained edge deployment. Notably, the newly released YOLOv12n exhibited a significant performance

Model	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)	Params/M
YOLOv8n	92.4	88.9	94.1	72.9	3.0
YOLOv10n	92.1	87.5	92.7	72.9	2.7
YOLOv11n	93.1	89.1	94.1	73.4	2.6
YOLOv12n	87.4	78.6	86.4	59.1	2.5
DETR	96.1	89.0	91.1	73.2	40
Faster R-CNN	95.3	86.7	90.2	–	32
YOLOv8n-CW	92.0	87.9	93.1	72.4	3.1
DSSKA-YOLOv8n	95.5	88.9	94.9	75.8	3.3

Table 4. Performance comparison of different algorithms.



Fig. 4. Detection results of DSSKA-YOLOv8n.

drop on our dataset. This highlights the need for task-specific optimization and validates the robustness of our approach. In summary, the comprehensive comparison demonstrates that DSSKA-YOLOv8n significantly enhances the detection accuracy for geological hazard targets while maintaining the efficient feature extraction characteristic of lightweight models. It achieves an excellent trade-off between accuracy and efficiency, providing a potentially valuable algorithmic option for efficient perception in edge devices such as rescue robots.

Detection result analysis of the proposed recognition algorithm

This section provides a comprehensive analysis of the detection performance, internal mechanisms, and limitations of the DSSKA-YOLOv8n model through qualitative and visualization methods. The detection results of the model for different categories of geological hazard targets are illustrated in Fig. 4. The figure includes four typical hazard types: rockfall, landslides, fallen_tree, and road_collapse. The model successfully achieves accurate detection and localization for all targets, visually validating its generalization capability for heterogeneous hazard types.

To highlight the effectiveness of the proposed DSSKA module, Fig. 5 provides a visual comparison of detection results with the baseline YOLOv8n model. The upper panel shows the detection results of the original YOLOv8n, which exhibits noticeable false positives and a failure to detect small-sized targets. In contrast, the



Fig. 5. Visual comparison before and after model improvements.

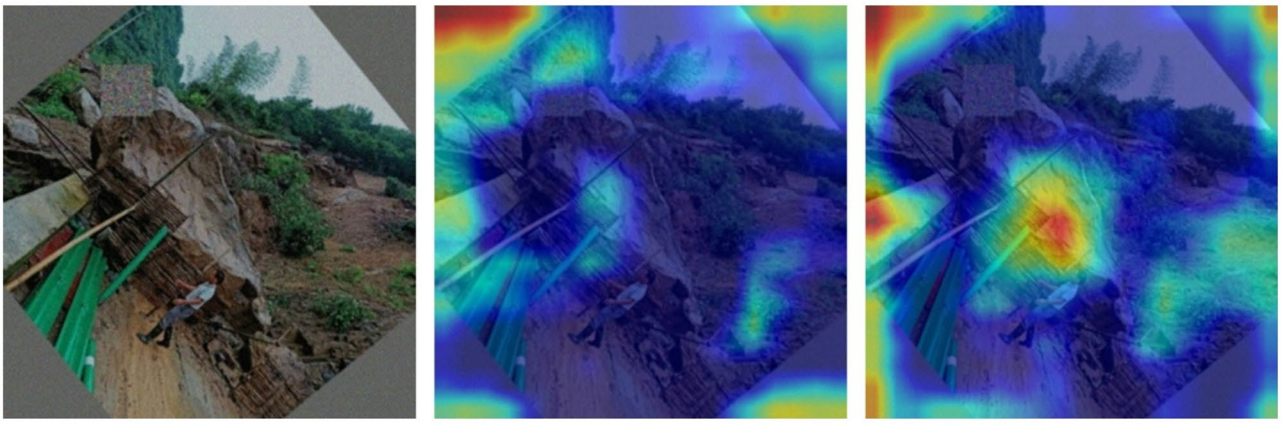


Fig. 6. Comparison of heatmaps before and after adding DSSKA.

lower panel demonstrates that, after embedding the DSSKA module, the model achieves more accurate and complete target identification under the same conditions. This effectively suppresses false positives, proving the crucial role of the proposed improvements in enhancing detection robustness in complex scenes.

To investigate the mechanism of the DSSKA module in rockfall detection, this study employs Grad-CAM to generate comparative heatmaps. Figure 6 shows that after incorporating DSSKA, the model exhibits enhanced response intensity to the boundaries and surface textures of the boulders themselves, with attention distribution becoming more concentrated on the central regions of the targets. However, the heatmaps also reveal that the model's attention weights are confounded at the boundaries between rockfall and shadows. This confusion correlates with the relatively lower mAP50-95 performance of 72.1% for the Rockfall category in Table 1, indicating that the current multi-scale design still has limitations in discriminative power for scenarios with ambiguous target-background boundaries.

Despite the significant performance improvement of the model, an analysis of the failure cases in Fig. 7 reveals three key limitations in certain scenarios. First, there is a lack of scene context awareness. As shown in the left and middle images of Fig. 7, a tunnel entrance and a mountain slope with yellowish-brown coloration were incorrectly identified as road collapses. The underlying cause is the model's failure to establish a spatial relationship modeling between targets and their surrounding scene attributes. Second, the feature representation capabilities for targets are overly simplistic, lacking constraints based on morphological and dynamic attributes. For instance, in the left image of Fig. 7, road lane markings were misdetected as rockfalls, and in the right image, a flooded river channel was misclassified as a rockfall. These errors stem from the model's inability to distinguish differences in shape and the dynamic characteristics of targets. Third, the capability to extract features of extremely small targets is insufficient. The current multi-scale design has limited capability in extracting features from tiny targets, which corresponds to the low detection confidence observed for some small-sized rockfalls. Addressing these limitations requires future improvements that address these core issues. First, introduce a context-aware module that models "target-scene" associations to strengthen the capability to match scene attributes with hazard targets. Second, expand the feature discrimination dimensions by adding dedicated branches for extracting attribute features such as morphology and state. Third, optimize the design of multi-scale convolution kernels by incorporating finer-scale branches specifically adapted for detecting extremely small targets[14].

Discussion

The core challenge for lightweight object detection models in geological hazard rescue scenarios lies in balancing the demand for high accuracy with limited computational resources. The DSSKA-YOLOv8n model proposed in this paper aims to address this challenge by introducing a Depthwise Separable Selective Kernel Attention module. Experimental results demonstrate a favorable balance: the model improves mAP50-95 by 2.9% over YOLOv8n with only a 10% parameter increase. Notably, under identical dataset and training configurations, DSSKA-YOLOv8n achieved a significantly higher mAP50-95 of 75.8% compared to the 72.4% of the YOLOv8n-CW model[6], which was also specifically designed for geological hazards detection. This further validates the effectiveness of the proposed lightweight attention mechanism in feature enhancement. Ablation studies confirm that this performance gain stems from the proposed lightweight DSSKA module. By reconstructing the SKA using depthwise separable convolutions, the module achieves a 0.7% increase in mAP50-95 while reducing parameters by 61.2%. Visualization evidence (Fig. 6) also indicates that the module enhances the focus of attention on key features such as the boundaries and textures of post-landslide rockfall.

However, the model has certain limitations alongside its positive results. Firstly, although data augmentation was applied, our custom dataset is still relatively limited in scope and geographically restricted to certain mountainous regions in Sichuan Province, China. While this ensures good model adaptation for this region, it may introduce a risk of overfitting and could potentially limit the model's generalization performance across different geological, geomorphological, and climatic conditions. Furthermore, an analysis of the model's capabilities, as revealed by the failure cases in Fig. 7 and the performance variations in Table 1, reveals specific limitations: First, the adaptive multi-scale design of the current DSSKA module still lacks sufficient granularity for capturing features of extremely small targets. Second, the model's ability to discriminate based on target morphology and state attributes is inadequate, exemplified by cases in Fig. 7 where white road markings or flood channels were misclassified as rockfalls. Third, there is scope for improving the model's robustness in scenarios involving vegetation occlusion. Finally, the validation in this study remains at the laboratory simulation stage, focusing on algorithm-level accuracy and lightweight evaluation. The model has not yet been deployed on a real-world rescue robot platform. Consequently, critical real-time performance metrics such as inference speed, power consumption, and system stability on real-world edge devices require empirical evaluation in subsequent work. Moreover, the current work is confined to pure visual perception and does not leverage the rich multi-source information available at hazard sites, such as acoustic and vibration data. Addressing these aspects constitutes a core set of problems to be solved in the future for developing more context-aware and fault-tolerant perception systems for rescue robots.



Fig. 7. Typical detection failure cases of DSSKA-YOLOv8n.

Although this study did not involve deployment on real edge devices, a preliminary feasibility analysis can be drawn based on the model's lightweight characteristics. With only 3.3M parameters, DSSKA-YOLOv8n exhibits a parameter scale comparable to other efficient architectures that have been successfully deployed on embedded platforms. For instance, recent studies have shown that after TensorRT optimization, lightweight YOLO-based models can achieve inference speeds exceeding 60 FPS on platforms such as the NVIDIA Jetson Xavier NX while maintaining competitive accuracy[15]. Furthermore, systematic benchmarks on Jetson Orin Nano have demonstrated that several YOLO variants with similar parameter counts can attain real-time performance (30–60 FPS) after proper optimization[16]. Given the lower parameter count and architectural efficiency of DSSKA-YOLOv8n, we infer that it has the potential to achieve > 30 FPS on similar edge devices, thereby meeting the preliminary real-time requirements for onboard perception in rescue robots. This projection, however, remains theoretical and will be empirically validated through actual deployment and benchmarking in our subsequent work.

In summary, the core contribution of DSSKA-YOLOv8n lies in providing a laboratory-validated, multi-scale adaptive visual perception algorithm for resource-constrained scenarios. This research validates the lightweight potential of combining depthwise separable convolutions with adaptive attention mechanisms. Future efforts should not only focus on optimizing mechanisms for small-target perception and attribute discrimination but also expand the dataset's coverage and scenario diversity, validate the model's real-time performance and energy efficiency on embedded platforms, and actively explore multi-modal information fusion. These steps aim to progressively build a rescue robot perception system with enhanced environmental understanding, decision-making fault tolerance, and practical deployment capability.

Methods

Experimental setup

All experiments were conducted under a unified hardware and software environment. The hardware platform consisted of an Intel Core i5-13600KF CPU and an NVIDIA GeForce RTX 4060 GPU (8GB VRAM). The software stack was built on the Windows 11 operating system, utilizing the PyTorch 2.0.0 deep learning framework, with CUDA 11.8 and cuDNN 8.7.0 for GPU acceleration.

For model training, we followed the standard configurations of the baseline YOLOv8n framework. The model was initialized with the official YOLOv8n weights pre-trained on the COCO dataset to accelerate convergence and improve generalization. The Stochastic Gradient Descent (SGD) optimizer was employed with a momentum of 0.937 and a weight decay factor of 5×10^{-4}. The initial learning rate was set to 0.01, and a cosine annealing scheduler was applied throughout the training process to adjust the learning rate dynamically. All input images were resized to 640×640 pixels. A consistent batch size of 8 was used across all models. Training was conducted for up to 1000 epochs, with an early stopping mechanism (patience of 50 epochs) monitoring the validation loss to prevent overfitting. Each experimental configuration was repeated five times to ensure the reliability of the reported results. All performance metrics presented in the tables are reported as mean values over these five independent runs. Owing to the fixed random seed of 42 used for all stochastic processes—including data partitioning, model initialization, and data augmentation—the standard deviations across runs were zero, guaranteeing full reproducibility of the experiments.

Dataset construction and augmentation

To ensure the effectiveness and generalization capability of model training, this study constructed a dedicated geological hazard dataset. The data were sourced from typical mountainous hazard-prone areas in Sichuan Province, China. Initial images were obtained by extracting frames from high-definition videos recorded on-site using cameras with a resolution of 1920×1080 pixels. A total of 2173 valid images were acquired, encompassing four categories of geological hazard targets: rockfall (740 images), fallen_trees (454 images), landslides (453 images), and road_collapse (526 images). Sample images are shown in Fig. 8. All images were meticulously annotated using the Labelme tool, with the annotation information consisting of the minimum bounding rectangles for the geological hazard targets and their corresponding category labels[17].

To enhance the model's robustness in complex field environments and address sample scarcity, a systematic data augmentation strategy was implemented. First, all images were uniformly cropped and resized to 640×640



Fig. 8. Example of dataset images.

Category	Before Augmentation	After Augmentation
ALL	2173	4346
Rockfall	740	1480
Landslides	453	906
Fallen_tree	454	908
Road_collapse	526	1052

Table 5. Sample distribution statistics before and after data augmentation.



Fig. 9. Comparison of dataset images before and after data augmentation.

pixels, and pixel normalization was applied to reduce variations in size and brightness from the original data. Subsequently, a multi-faceted augmentation strategy was employed to simulate real-world perturbations: this included random rotation and color jittering to improve the model's adaptability to viewpoint and lighting variations; and the addition of random Gaussian blur and rectangular masking blocks to simulate image degradation caused by rainy/foggy weather and partial occlusion of targets by dust or vegetation. These augmentations enhance the model's recognition capability under adverse conditions[18]. It is important to note that the original dataset exhibited a mild class imbalance, which could easily lead the model to overfit the majority classes and insufficiently learn features of minority classes. Therefore, we applied the aforementioned augmentation techniques to effectively double the total number of images for each category, resulting in a balanced dataset. This approach mitigated the bias introduced by the imbalanced distribution while avoiding the overfitting problems associated with simple oversampling. Following this augmentation process, the total dataset size was expanded to 4346 images. The detailed class distribution is presented in Table 5. A comparison of images before and after augmentation is shown in Fig. 9. Finally, the dataset was partitioned at the image level via stratified random sampling, dividing it into training, validation, and test sets in an 8:1:1 ratio. This ensured that the class distribution across subsets remained similar to the original dataset, preventing partitioning bias from interfering with model evaluation.

Evaluation metrics

To comprehensively evaluate the model performance, this study adopted Precision, Recall, mean Average Precision (mAP), and parameter count (Params) as the evaluation metrics.

Precision measures the proportion of correctly predicted positive targets; high precision reduces false positives, and its formula is given in Eq. (1):

$$Precision = \frac{TP}{TP + FP} \tag{1}$$

where TP (True Positive) is the number of positive samples correctly predicted as positive, and FP (False Positive) is the number of negative samples incorrectly predicted as positive.

Recall is the ratio of correctly predicted positive samples to the total number of positive samples, as shown in Eq. (2):

$$Recall = \frac{TP}{TP + FN} \tag{2}$$

where FN (False Negative) denotes the number of positive samples incorrectly classified as negative.

mAP is the mean of Average Precision (AP) across all categories, as shown in Eq. (3):

$$mAP = \frac{1}{n}\sum_{i=1}^{n}\int_{0}^{1} P_i(R)dR \tag{3}$$

where n is the total number of categories, i is the category index (ranging from 1 to n), R denotes recall, $P_i\,(R)$ is the precision as a function of recall R for the i -th category, and $\int_0^1 P_i(R)dR$ is the AP for that category.

mAP50 is the mean Average Precision at an IoU threshold of 0.5. mAP50-95 is a more stringent metric that averages mAP over IoU thresholds from 0.50 to 0.95 in steps of 0.05.

The number of parameters (Params) indicates the model size and complexity. In the field of deep learning, the number of parameters is usually one of the indicators to measure the size and complexity of the model, and a larger number of parameters usually means that more computational resources are required to train and run inference.

DSSKA-YOLOv8n algorithm

To address the core challenges of extreme scale heterogeneity, weak discriminative features of geological hazard targets, and the need for compatibility with lightweight platforms such as rescue robots, this paper proposes the DSSKA-YOLOv8n algorithm based on the YOLOv8n framework. Its core innovation lies in embedding a lightweight Depthwise Separable Selective Kernel Attention module at the terminal stage of the backbone network, with the overall architecture illustrated in Fig. 10. This design aims to enhance the model's perception of multi-scale targets with weak features at a minimal computational cost. The algorithm is elaborated as follows:

(1) Feature Enhancement in the Backbone Network: The feature extraction follows the CBS module (Conv-BN-SiLU) and the C2f. residual structure of YOLOv8n. A DSSKA module is embedded after the final C2f. stage. It employs multi-scale (1×1, 3×3, 5×5, 7×7) depthwise separable convolutions to adaptively capture cross-scale features, followed by an SPPF module to expand the receptive field. This design enhances the model's representational capacity for multi-scale geological hazard targets without significantly increasing the computational burden.
(2) Feature Fusion Network: The cross-scale fusion topology of YOLOv8n's PAN-FPN is retained. Multi-resolution feature aggregation is achieved through upsampling and concatenation operations, followed by semantic refinement via C2f. modules. This mechanism leverages the low-level features enhanced by the preceding DSSKA module, improving both the preservation of small-scale details and the effectiveness of large-scale semantic fusion. This enhances target distinctiveness against complex topographic backgrounds.
(3) Detection Head and Lightweight Design: The native YOLOv8n dual-branch detection head architecture is employed. It outputs localization and classification predictions through parallel convolutional branches, achieving localization and category determination for geological hazard targets via separate optimization of localization and classification via a joint loss function. The lightweight nature of DSSKA keeps the overall parameter count close to that of native YOLOv8n, thereby supporting its deployment feasibility on resource-constrained edge devices.

In summary, DSSKA-YOLOv8n achieves targeted enhancement of the native YOLOv8n's multi-scale feature extraction capability through the strategic integration of the DSSKA attention module. Experimental results indicate that this improvement effectively boosts geological hazard detection accuracy while maintaining a moderate parameter increase. This lays a technical foundation for the subsequent engineering deployment and integration of the model on rescue robot platforms.

DSSKA attention mechanism

In the initial design phase, we explored a more complex attention module named CSDSKA (Cross-Scale interactive Dynamic Selective Kernel Attention), which incorporated cross-scale interaction and a dynamic temperature coefficient. However, based on ablation studies (Tables 2 and 3), we found that a simplified version focusing solely on depthwise separable convolutions achieved superior performance with lower computational overhead. Specifically, the removal of the cross-scale interaction and dynamic temperature components reduced model complexity while improving detection accuracy (mAP50-95) by 0.7% compared to the standard SKA module, as shown in Table 2. Table 3 further validates the optimal deployment of DSSKA within the backbone network. Therefore, we adopted DSSKA as the final lightweight core design.

The core objective of integrating attention mechanisms into object detection architectures is to guide the model to dynamically focus on task-critical features while suppressing redundant background interference, thereby enhancing detection robustness in multi-scale and complex scenes. To address the specific requirements of geological hazard detection, this paper systematically evaluated several mainstream attention mechanisms (comparative results are shown in Table 6). Experiments indicate that the SKA (Selective Kernel Attention)

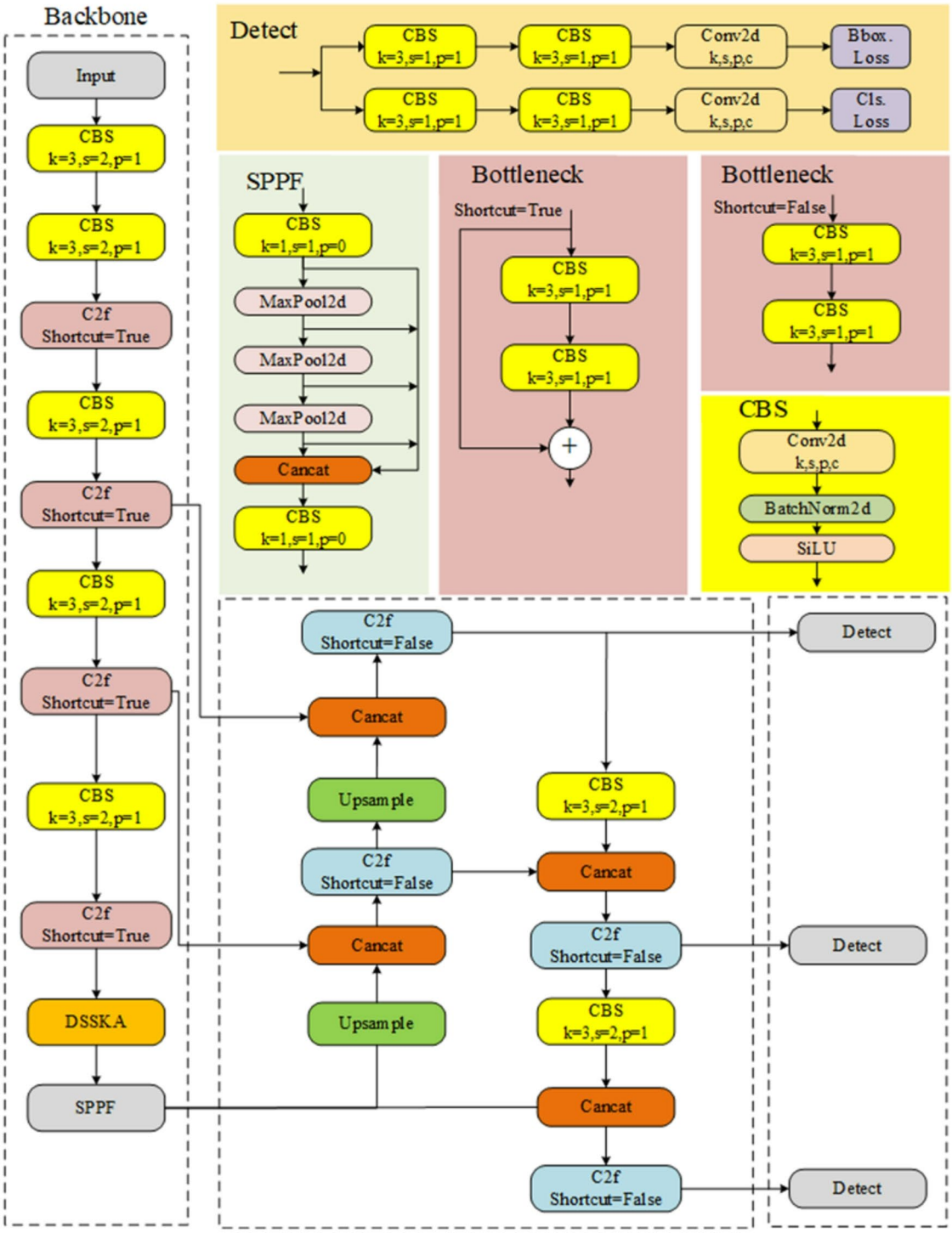


Fig. 10. Structure of DSSKA-YOLOv8n.

mechanism performs best in multi-scale feature capture (mAP50-95: 75.1%). However, its parameter count (8.5M) is substantially high—approximately 1.6 times that of the CA (Channel Attention) mechanism—highlighting a pronounced trade-off between accuracy and lightweight design[19].

To balance the requirements of accuracy and lightweight design, this paper proposes the DSSKA attention mechanism, which achieves a lightweight redesign of SKA through the following key improvements: (1) Lightweight Multi-scale Feature Extraction: The standard convolutions in SKA are replaced with a depthwise separable convolution architecture. Parallel branches employing 1×1, 3×3, 5×5, and 7×7 kernels, each followed by batch normalization and the SiLU activation function, extract multi-scale features. This design significantly reduces the parameters and computational cost of constructing the multi-scale branches while retaining cross-

Model	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)	Params/M
Yolov8n-SKA	94.3	89.5	94.2	75.1	8.5
Yolov8n-AWGAM	93.5	88.8	94.2	74.1	4.8
Yolov8n-CA	93.8	89.0	94.3	74.5	5.2
Yolov8n-PSA	94.1	89.2	94.2	74.8	6.5

Table 6. Comparison of different attention mechanisms.

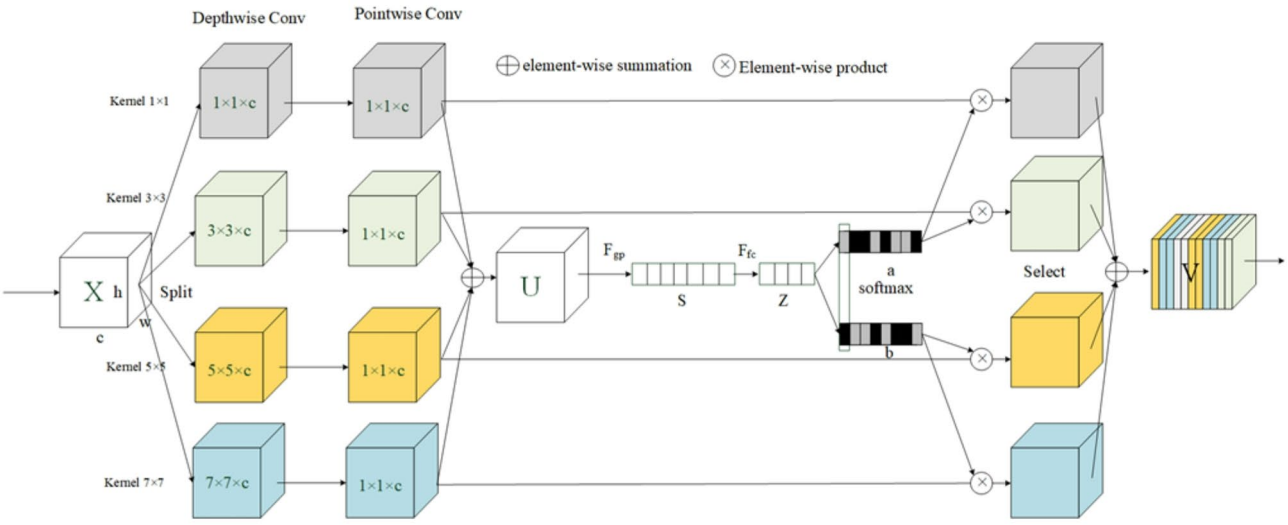


Fig. 11. Structure of DSSKA.

Model	Precision (%)	Recall (%)	mAP50(%)	mAP50-95(%)	Params/M
SKA	94.3	89.5	94.2	75.1	8.5
DSSKA	95.5	88.9	94.9	75.8	3.3

Table 7. Performance comparison between SKA and DSSKA.

scale representational capacity, aligning with the resource constraints of edge devices. (2) Adaptive Spatial Weight Allocation: The multi-scale features are stacked, followed by an adaptive weighted fusion across scales. Specifically, spatial dimensions are compressed via global average pooling. Spatially-adaptive scale weights are then generated through fully connected layers, normalized by Softmax, and used for a weighted summation of the feature maps from each scale. This mechanism enables the model to dynamically adjust the receptive field based on spatial location—enhancing fine-grained feature weights for small target regions and emphasizing global context for large target areas. This represents a key advancement over SKA, shifting from globally shared scale weights to spatially adaptive scale selection. The structure of DSSKA is illustrated in Fig. 11, and its workflow corresponds to the steps shown in the figure: multi-scale feature extraction → stacking and integration → scale-wise summation → global pooling → feature compression → weight generation → weighted fusion.

To validate the effectiveness of the lightweight design, we compared the performance of DSSKA with the original SKA (Table 7). DSSKA successfully achieved a synergistic optimization of accuracy and lightweight design: it significantly reduced the parameter count by 61.2% to 3.3M while simultaneously improving the core detection metric (mAP50-95) by 0.7%.

Depthwise separable convolution decouples traditional convolution into two stages: Depthwise Convolution and Pointwise Convolution. The former independently extracts spatial features for each input channel, while the latter achieves cross-channel information interaction via 1×1 convolutions. This hierarchical calculation not only significantly reduces the model's parameter count and computational complexity but also enhances the network's generalization ability by reducing parameter redundancy, effectively mitigating the risk of overfitting. In addition, the computational load of depthwise separable convolution has a linear relationship with its channel count. This allows the model to flexibly adjust structural complexity based on different computational resource requirements. This feature has been verified to be compatible with resource-constrained devices in edge diagnostic scenarios for industrial equipment[20]. Experimental results demonstrate that the improved DSSKA module maintains detection precision while achieving a substantial reduction in parameter count. It

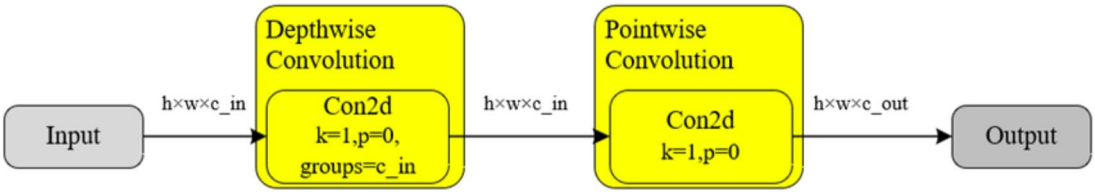


Fig. 12. Schematic diagram of the DepthwiseSeparableConv principle.

simultaneously enhances the model's lightweight capability and inference efficiency. The principle of depthwise separable convolution is illustrated in Fig. 12.

From a geomorphological perspective, the multi-scale adaptive receptive field of DSSKA mimics the expert interpretation logic of "from whole to part" and "from macro-morphology to micro-texture". In landslide identification, for instance, interpreters first delineate overall boundaries based on macro-scale features such as scarps and bulging toes—key indicators of landslide extent and deformation stage. They then focus on internal details at finer scales, including tension cracks and surface roughness, which reflect ongoing instability and potential reactivation[21]. This multi-scale reasoning aligns with recent findings that integrating multi-scale topographic attributes such as slope, texture, and topographic position significantly outperforms single-band visualizations in CNN-based geomorphic feature extraction[22]. DSSKA embodies this reasoning through multi-scale depthwise separable convolutions that capture features across varying receptive fields, while its adaptive spatial weight allocation dynamically emphasizes fine-grained textures like tension cracks and integrates global context represented by landslide boundaries. This design thus enhances the model's sensitivity to critical geomorphic indicators for distinguishing hazards from natural terrain.

Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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Author contributions

K.R. and X.X. designed the research and performed experiments; J.M. and D.G. processed data and analyzed results; G.S. contributed to the design of the DSSKA attention mechanism, analyzed experimental results, and supervised the project; J.N. prepared the figures and tables; J.Y. conducted literature investigation; J.S. conceived the study, wrote the main manuscript text, and supervised the project. All authors reviewed and revised the manuscript.

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Declarations

Competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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